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Optimal time-domain technique for pulse width modulation in power electronics

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Optimal time-domain technique for pulse width modulation is presented. It is based on exact and explicit analytical solutions for inverter circuits, obtained for any sequence of input voltage rectangular pulses. Two optimal criteria are discussed and illustrated by numerical examples. © *2018 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (http://creativecommons.org/licenses/by/4.0/).* https://doi.org/10.1063/1.5006468

I. INTRODUCTION

The pulse width modulation (PWM) is widely used in power electronics for the operation of inverters.[1,2] Inverters are essential in the design of uninterruptible power supplies (UPS), in integrating renewable energy sources into existing power systems and for frequency control of speed of motors (semiconductor drives). In particular, they are used for speed control of spindle motors of hard-drives in magnetic recording. Currently, frequency domain techniques based on Fourier series expansion of rectangular pulse sequences (trains) are used for the design of various PWM schemes. The central idea of these frequency domain techniques is to suppress lower order sinusoidal harmonics, at the expense of the enhancement of higher-order harmonics. The latter harmonics are usually suppressed by the presence of inductors in inverter circuits.

In this paper, a new optimal time-domain technique for the development of PWM is presented and illustrated by numerical examples. This time domain technique is based on exact analytical solutions to electric circuit equations of inverters that can be found for any sequence (any train) of rectangular pulses. These analytical solutions are explicit functions of time instants which specify the beginnings and ends of individual rectangular pulses. For this reason, these analytical solutions can be used for the optimal design of PWM schemes. Namely, the aforementioned time instances can be selected to minimize the chosen criteria for the deviations of the above-mentioned analytical solutions from desired sinusoidal currents (sinusoidal output voltages) of inverters. Below, two criteria are discussed. The first one is the least-square criteria which results in the minimization of the total content of higher-order harmonics. The second one is the constrained least-square criterion which guarantees the elimination of certain harmonics and the minimization of the total content of the remaining higher-order harmonics.

II. TECHNICAL DISCUSSION

Consider the single-phase bridge inverter as shown in Figure 1(a). Here, SW_1, SW_2, SW_3 and SW_4 are bidirectional (bilateral) switches which are usually realized as parallel connections of transistors and free-wheeling diodes. These switches are used to generate a voltage between nodes 1 and 2 which is a train of rectangular pulses of height V_o. This means that the circuit in Figure 1(a) can be replaced by the equivalent circuit shown in Figure 1(b), where the equivalent voltage source $v_{eq}(t)$ is as shown

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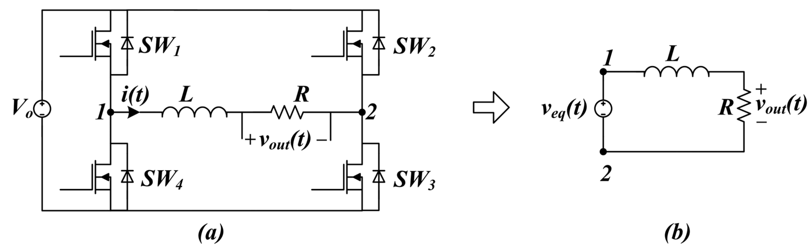

FIG. 1. (a) Circuit Diagram of Single-Phase Inverter, (b) Equivalent Circuit Representation.

in Figure 2. This equivalent voltage source is a rectangular pulse train with the half-wave symmetry. For this reason, we consider only the time interval 0 < t < T/2.

It is clear that:

$$v_{eq}(t) = \begin{cases} 0, & \text{if } t_{2j} < t < t_{2j+1}, \\ V_o, & \text{if } t_{2j+1} < t < t_{2j+2}, \end{cases} \tag{1}$$

where $j = 0,1,\ldots,k$ and $t_o = 0$, while $t_{2k+1} = T/2$.

From Equation (1) and Figure 1(b) we find:

$$i(t) = \begin{cases} A_{2j+1} e^{-\frac{R}{L}t}, & \text{if } t_{2j} < t < t_{2j+1}, \\ A_{2j+2} e^{-\frac{R}{L}t} + \dfrac{V_o}{R}, & \text{if } t_{2j+1} < t < t_{2j+2}, \end{cases} \tag{2}$$

where the constants A_i (for $i = 1,2,3,\ldots,2k$) are found from the continuity condition for electric current $i(t)$ at times t_{2j} and t_{2j+1} as well as from the half-wave symmetry boundary condition

$$i(0) = -i(T/2). \tag{3}$$

The above conditions result in linear simultaneous equations for the A-coefficients with a two-diagonal matrix. The solution of these equations yields the following analytical expressions for the coefficients:

$$A_1 = -\frac{V_o}{R} \cdot \frac{\sum_{j=1}^{2k} (-1)^j e^{\frac{R}{L}t_j}}{1 + e^{\frac{-RT}{2L}}} \cdot e^{\frac{-RT}{2L}} \tag{4}$$

$$A_{2k+1} = \frac{V_o}{R} \cdot \frac{\sum_{j=1}^{2k} (-1)^j e^{\frac{R}{L}t_j}}{1 + e^{\frac{-RT}{2L}}} \tag{5}$$

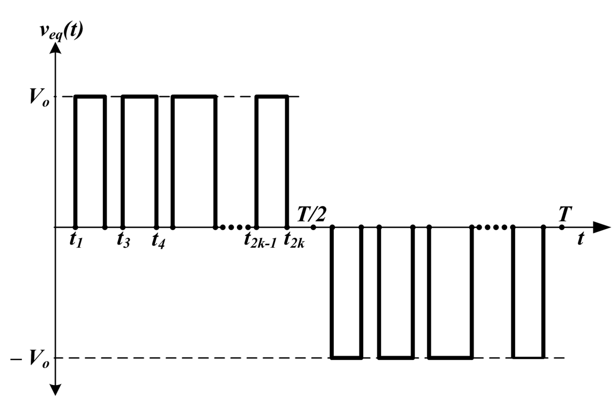

FIG. 2. Equivalent voltage (v_{eq}) waveform.

$$A_j = A_1 + \frac{V_o}{R}\sum_{n=1}^{j-1}(-1)^n e^{\frac{R}{L}t_n} \tag{6}$$

The last three equations along with (2) provide the exact, analytic expression for $i(t)$ in terms of time instants t_j:

$$i(t,\ t_1,\ t_2, \ldots, t_{2k}). \tag{7}$$

For this reason, the problem of optimal design of PWM can be stated as the following optimization problem: find the times $t_1,\ t_2,\ ....,\ t_{2k}$ such that the function (2) gives the best (in some sense) approximation to the desired function

$$\tilde{i}(t) = I_{ma} sin(\omega t - \phi), \tag{8}$$

where

$$\omega = \frac{2\pi}{T} \quad \text{and} \quad \tan\phi = \frac{\omega L}{R}. \tag{9}$$

In particular, the least square criteria can be chosen for the selection of t_j. In this case, the problem is reduced to the minimization of the following integral:

$$E_2(t_1, t_2, \ldots, t_{2k}) = \int_0^{T/2} \left[\ i\ (t, t_1, t_2, \ldots, t_{2k}) - \tilde{i}(t)\right]^2 dt. \tag{10}$$

According to Parseval's theorem, $E_2(t_1,t_2,\ldots,t_{2k})$ is directly related to the sum of squares of the peak-values of all time harmonics of the integrand in (10). In this sense, $E_2(t_1,t_2,\ldots,t_{2k})$ can be interpreted as the total harmonic content, and the optimal PWM results in the minimization of this total higher order harmonic content.

Another practically meaningful optimization approach is to minimize the integral in (10) subject to the specific constraints which guarantee the elimination of certain lower-order harmonics in $i(t,\ t_1,\ t_2,\ldots,t_{2k})$. These constraints can be derived from the conditions that corresponding Fourier coefficients of $v_{eq}(t)$ (presented in Figure 2) are made equal to zero. In particular, if the m^{th} harmonic is to be eliminated, the following constraint needs to be satisfied:

$$\sum_{j=1}^{2k}(-1)^{j+1} cos m\omega\ t_j = 0. \tag{11}$$

Similarly, the constraint can be imposed that the first harmonic of $i(t,\ t_1,\ t_2,\ \ldots,\ t_{2k})$ exactly coincides with $\tilde{i}(t)$ given in equation (8) and (9). The mathematical form of this constraint is as follows:

$$\sum_{j=1}^{2k}(-1)^{j+1} cos\omega\ t_j = \frac{\pi\ I_{ma}\sqrt{R^2 + (\omega L)^2}}{2V_o}. \tag{12}$$

The minimization of integral (10) subject to constraints (11) and (12) will result in PWM that eliminates selected harmonics due to the constraints (11), exactly produces the desired fundamental component $\tilde{i}(t)$ given in equation (8) and (9), and minimizes the total content of the remaining higher-order harmonics.

The minimization of the integral (10) can be reduced to the minimization of explicit function $E_2(t_1,t_2,\ldots,t_{2k})$. This function is derived by substituting formulas (2) and (8) into equation (10) and subsequently performing the integration. The outlined tedious derivation leads to a lengthy analytical expression for $E_2(t_1,t_2,\ldots,t_{2k})$, which is omitted here. The minimization of this function can then be performed under the constraint that the values of times $t_1,\ t_2,\ldots,\ t_{2k}$ should be monotonically increasing. The latter can be enforced by introducing new variables:

$$\Delta t_n = t_n - t_{n-1}; \qquad n = 1, 2, \ldots, 2k, \tag{13}$$

and by expressing the function to be optimized, as well as the constraints (11) and (12) in terms of Δt_n using the formulae:

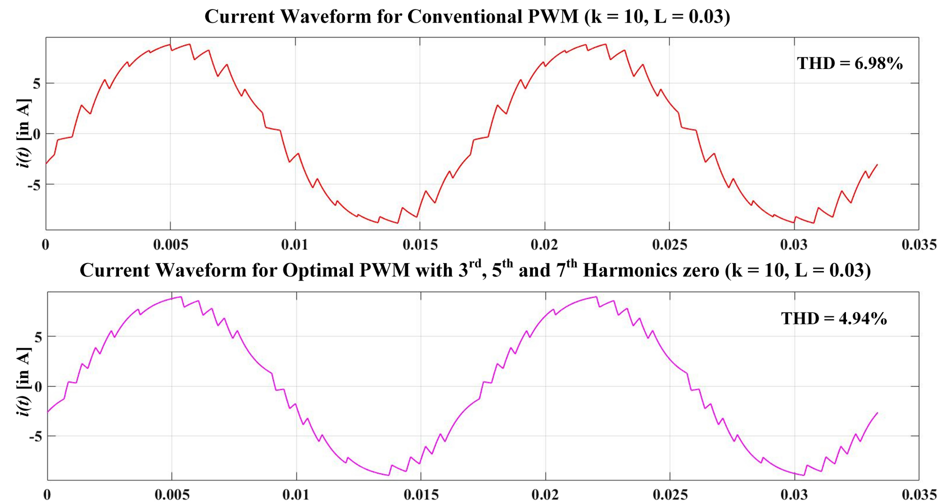

FIG. 3. Current waveform for Single Phase H-Bridge Inverter.

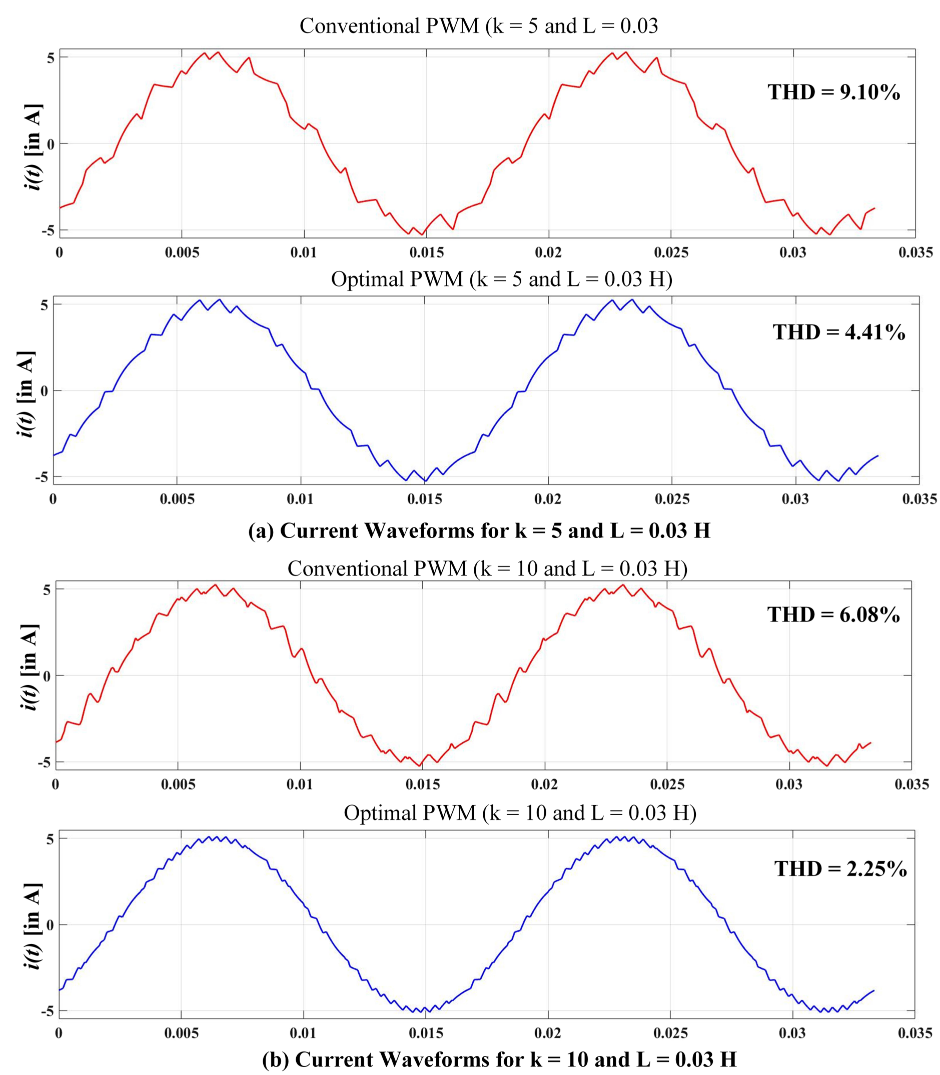

FIG. 4. Current Waveforms for three-level NPC single-phase inverters for L = 0.03 H and (a) k = 5 (b) k =10.

$$t_1 = \Delta t_1 \qquad \text{and} \qquad t_j = \sum_{n=1}^{j-1} \Delta t_n. \tag{14}$$

Then, the region of the minimization of the corresponding function $E_2(\Delta t_1, \Delta t_2, \ldots, \Delta t_{2k})$ is reduced to the convex region (to the cone) specified by the inequalities:

$$\Delta t_n > 0 \qquad \text{and} \qquad \sum_{\mathrm{n=1}}^{2k} \Delta t_n < \frac{T}{2}. \tag{15}$$

In the performed calculations, the interior-point method,[3,4] as well as the sequential quadratic programming (SQP) technique[5] were used to minimize the function $E_2(\Delta t_1, \Delta t_2, \ldots, \Delta t_{2k})$ subject to the desired constraints. Software implementations of these techniques are readily available in MATLAB. These techniques produced almost identical results. However, the SQP technique exhibited monotonic convergence, while the convergence of the interior point method was not always monotonic.

The presented analysis can be extended to circuits more complex than the R-L circuit in Figure 1(b). This extension is based on the notion of unit step response and the superposition integral for electric circuits. Furthermore, the optimal time-domain PWM technique is presented above for the case of a single-phase inverter. It can be generalized to the case of three-phase inverters. However, due to the lack of space, these matters are left unattended in this paper. Instead, in conclusion, we illustrate the described technique by some numerical examples and compare it with the conventional frequency-domain PWM technique, where the rectangular pulses are centered at the time instants:

$$t_p = \frac{T}{2k}\left(p + \frac{1}{2}\right), \qquad \text{where } p = 0, 1, \ldots, (k-1) \tag{16}$$

and their width is modulated according to formulae:

$$\Delta t_p = \frac{mT}{2k} sin\omega t_p, \qquad \omega = \frac{2\pi}{T} \quad \text{and } 0 < m < 1. \tag{17}$$

For comparison purposes, we performed the calculations for conventional PWM and optimal time-domain PWM techniques in the cases of single-phase H-Bridge inverter as well as for 3-level Neutral Point Clamped (NPC) single-phase inverter.[1] These computations are presented in Figure 3 and Figure 4 respectively.

The presented results clearly reveal the more effective harmonic distortion suppression by using the optimal time-domain PWM technique. The optimal time-domain PWM technique can completely eliminate selected higher order harmonics. The latter is not strictly possible within the framework of the conventional frequency-domain PWM technique.

It is worthwhile to remark that parallel implementation of optimization algorithms by using GPUs (Graphics Processing Unit) has been developed.[6,7] This implementation may substantially speed up realization of optimal time-domain PWM and can be used for generation of PWM pulses in real-time. Alternatively, the optimization can be performed offline for different values of the maximum current, $I_{ma,}$ and the values of the corresponding optimal time-instants can be stored in a look-up table (LUT) for implementation in real-time. This method is widely popular for implementation of PWM for inverters.[1,8,9]

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